

M.R. JAYASURIYA MUTHUKUMAR

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PROFESSIONAL SUMMARY

Dedicated and experienced Service Engineer with over 10 years of expertise in installation, troubleshooting, preventive maintenance, and repair of industrial and commercial diesel generators. Proficient in electrical systems, control panels, load bank testing, and HVAC integration. Adept at reducing downtime, improving system efficiency, and ensuring safety and compliance across all tasks. Strong record of technical mentorship and teamwork.

CORE COMPETENCIES

- Electrical Troubleshooting & Schematic Analysis
- Diesel & Gas Engine Diagnostics
- Generator Installation & Preventive Maintenance
- Load Bank Testing & Control Panel Repair
- HVAC System Integration & Plumbing Basics
- Technical Documentation (CMMS) & Reporting
- Customer Service & Team Collaboration

PROFESSIONAL EXPERIENCE

S.P.B Engineering works pvt ltd – Trainee Technician

2012 - 2014

S.P.B Engineering works pvt ltd - Senior Generator Service Engineer

2017 - Present

- Performed preventive maintenance on over 100+ industrial diesel generators, reducing downtime by upto 30%.
- Diagnosed and repaired complex faults related to fuel systems, control panels, and electrical connections.
- Led the installation and commissioning of more than 8 backup power systems, including automatic transfer switches.
- Trained and mentored junior technicians, improving onboarding efficiency and technical capabilities.

- Ensured strict compliance with safety protocols and OEM standards during all service activities.

ELECTRICAL SKILLS

- Electrical fault diagnosis and troubleshooting
- Alternator winding inspection and repair
- Generator control panel wiring and configuration
- AVR (Automatic Voltage Regulator) installation and calibration
- Load testing and electrical performance analysis
- Battery charging system testing and replacement
- Voltage and frequency regulation adjustment
- Electrical safety procedures and lockout/tagout (LOTO)
- Cable termination, insulation testing, and continuity checks
- Synchronization of multiple generators (parallel operation)
- Circuit breaker installation and maintenance
- Earth/grounding system inspection and testing
- Electrical schematic reading and interpretation
- Preventive electrical maintenance scheduling

ELECTRONIC SKILLS

- Electronic control module (ECM) troubleshooting and programming
- Digital generator control panel setup and calibration
- Automatic Transfer Switch (ATS) installation and testing
- Sensor testing and replacement (temperature, oil pressure, RPM, etc.)
- AVR (Automatic Voltage Regulator) electronic adjustment and replacement
- Synchronization control system setup (for parallel generators)
- Electronic fuel injection (EFI) system diagnostics
- Fault code reading and reset using diagnostic tools
- PLC-based generator control integration
- Electronic safety interlock systems maintenance
- Monitoring and configuration of remote generator management systems
- Frequency regulation via electronic governors
- Voltage and power factor correction using electronic modules
- Testing and repair of PCB boards in generator controllers
- Installation of monitoring devices for load and runtime tracking

TOOLS & TECHNOLOGIES

- Electrical Testing Tools: Multi-meter (Fluke), Oscilloscope, Load Bank Testers
- Software: Microsoft Office Suite, CMMS Systems, Basic CAD
- Technical Knowledge: Three-phase Systems, Wiring Diagrams, OEM Manuals
- Brands Worked With: Cummins, Kirloskar, Greaves, Caterpillar

Technical Projects & Achievements

- Generator Load Testing: Led full load tests on 200 kW units, verifying voltage/frequency stability under stress conditions per ANSI standards.
- Control System Optimization: Upgraded generator control panels with AVRs and remote monitoring, improving performance reliability by 30%.
- Downtime Reduction: Implemented proactive maintenance schedules cutting downtime by 20% and saving.

ACADEMIC QUALIFICATIONS

Course	Institution	University Board	Year Of Passing	Percentage
Diploma in Automobile Engineering	Krishnasamy Memorial Polytechnic College, Tamil Nadu	Anna University, Chennai	2014 - 2017	68%
SSLC	Lakshmi Chordia Matriculation Memorial School, Cuddalore.	State Board	2012	60%

PROFESSIONAL DEVELOPMENT

- Hands-on experience with OEM-certified equipment (Cummins, Kirloskar, Greaves, Caterpillar)
 - Continuous self-learning through manuals, field mentoring, and on-the-job training
 - Strong understanding of interdisciplinary knowledge: electrical, mechanical, and plumbing systems
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DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I take full responsibility for the correctness of the information mentioned in this resume.

Date: _____

Place: _____

Signature: _____